Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/03/2007
PI Name:	Shelby, James Ph.D.		
Project Title:	Hydrogen Loaded Hollow Glass Microspheres as a Multi-Functional Radiation		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	09/01/2004	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	DeRosa, Rebecca (Alfred University) Hall, Matthew (Alfred University)		
Grant/Contract No.:	NNM04AA58G		
Performance Goal No.:			
Performance Goal Text:			
	The proposed project will investigate the feasibility of employing hollow glass microspheres (HGM) filled with hydrogen gas to stop heavy energetic ions. In this case, the HGM act as miniature pressure vessels capable of containing hydrogen gas at a pressure on the order of 10,000 psi. The hydrogen is loaded by heating the HGM in a vessel containing hydrogen, which then diffuses through the thin wall of the HGM and into the hollow cavity. The hydrogen is retained by returning the loaded HGM to ambient temperature conditions so that outward diffusion is substantially reduced. The use of HGM for hydrogen storage has several advantages. HGM production is a mature, well-established technology that is capable of producing substantial quantities of product from relatively inexpensive materials that are easily recyclable. Hydrogen storage in HGM is also extremely safe since the total amount of gas contained within any single sphere is quite minute. Therefore, in the event that a HGM should fracture, only a small quantity of hydrogen		

Task Description:	would be released. As an extension of the hydrogen-filled HGM concept, we propose to investigate the manufacture of HGM-polyethylene composite materials. The shielding properties of polyethylene are well documented, but its use in a composite provides additional benefits. First, the polyethylene matrix would act as a convenient storage medium for the hydrogen-filled HGM that act as freely flowing powder. Second, the thermoplastic properties of polyethylene could potentially be used to produce composite materials in a variety of geometries including tiles, flexible sheets, and complex form-fitting parts. Finally, it should be noted that the inclusion of hydrogen-filled HGM in a polyethylene matrix would produce a composite material with a hydrogen density equal to or greater than polyethylene, but at a fraction of the weight. The proposed work will employ two graduate students. One student will explore the production of HGM-polyethylene composite materials. The influence of HGM loading on the rheology and processing of polyethylene melts will be investigated. The strength of the composite materials and the HGM/polyethylene interface will be studied using the mechanical testing facilites available at Alfred University. A second student will study the radiation shielding properties of isolated hydrogen-filled HGM as well as HGM-polethylene composite materials, presumably using the facilities at Brookhaven National Laboratory. The study will initially employ standard commercially available HGM, but the development of alternate HGM compositions will be carried out by the student in conjunction with MoSci, Inc. Compositional development will focus on glasses containing substantial quantities of lithium and boron, since both elements offer the additional functionality of a large neutron absorption cross-section. Relevant issues that must be addressed include the adaptation of the traditional HGM production process to lithium borate glasses and the capacity of lithium borate glasses to imbibe and subsequently retain hydrogen gas. In addition, the possibility of radiation-induced defects in the glass will be investigated using electron paramagnetic resonance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The technology explored during the research is intended for use as radiation shielding in spacecraft subjected to galactic cosmic radation, and so benefit to life on Earth is minimal. Nevertheless, the subject of hollow glass microsphere/polyethylene composites is of interest for other Earth-based applications (e.g., light weight/buoyant materials). The research also investigated novel gadolinium-containing borate glasses that may be of general use for radiation shielding applications.
Task Progress:	Prior to cancellation of the radiation shielding materials program, we successfully prepared and tested hollow glass microsphere/polyethylene composites and tested their density and mechanical strength. Extensive work was also performed on the formation of novel hollow glass microspheres from borate-based glasses. This aspect of the work was not successful, most likely due to the viscosity characteristics of the borate glasses -- that is, the temperature-dependence of the borate glass viscosity was not amenable to hollow microsphere formation. Structure/property relationships in the borate glasses were further clarified using Raman spectroscopy, X-ray photoelectron spectroscopy, EPR spectroscopy, and viscosity modeling using results from Differential Scanning Calorimetry.
Bibliography Type:	Description: (Last Updated: 04/03/2007)
Articles in Peer-reviewed Journals	Geleil AS, Hall MM, Shelby JE. "Hollow glass microspheres for use in radiation shielding." Journal of Non-Crystalline Solids. 2006 May 15;352(6-7): 620-5. http://dx.doi.org/10.1016/j.jnoncrysol.2005.11.049 , May-2006
Articles in Peer-reviewed Journals	Ashton-Patton MS, Hall MM, Shelby JE. "Formation of low density polyethylene/hollow glass microsphere composites." Journal of Non-Crystalline Solids. 2006 May 15;352(6-7):615-9. http://dx.doi.org/10.1016/j.jnoncrysol.2005.11.058 , May-2006